DataCo
SUPPLY CHAIN ANALYTICS

Executive Overview

FILTERS

Year: 2015 | 2016 | 2017 | 2018

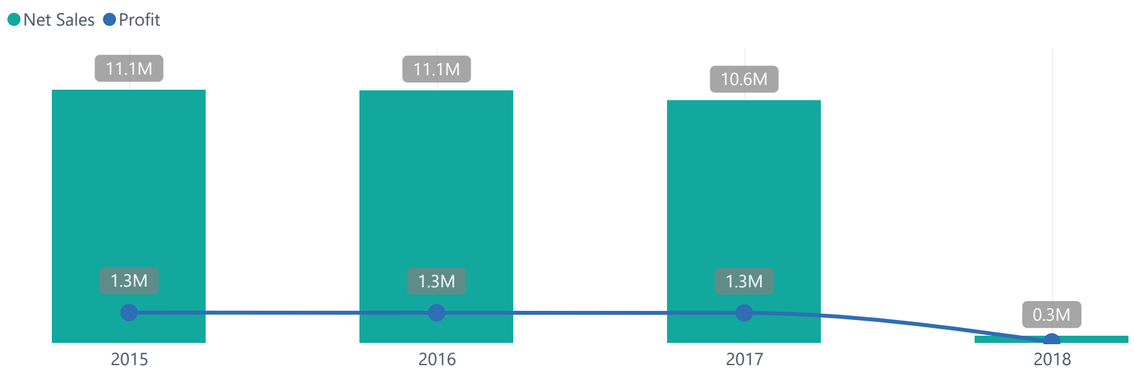
Shipping Mode: All

Customer Segment: All

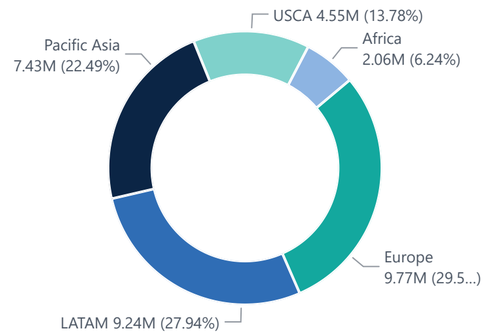
Market: All

33.05M	3.97M	12%	66K	57.29%	18.71%
Sales	Profit	Profit Margin %	Total Orders	Late Delivery %	Loss Making Items %

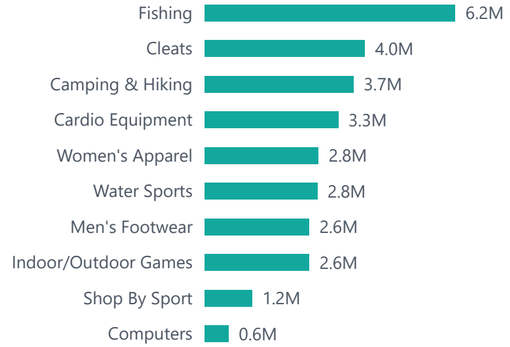
Sales and Profit by Year/Month



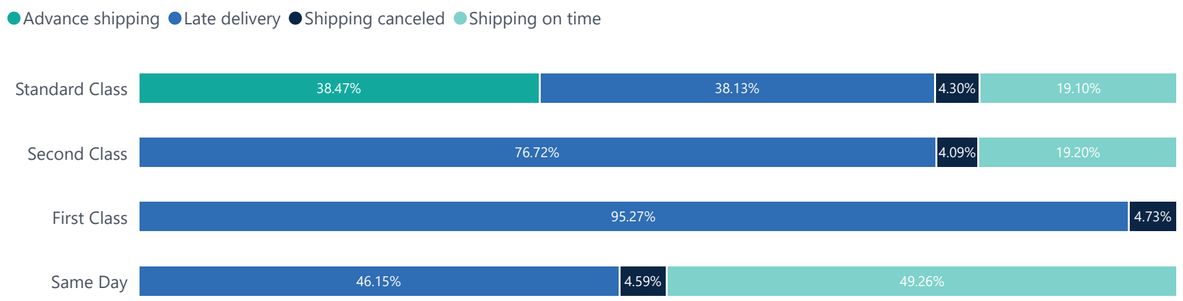
Sales by Market



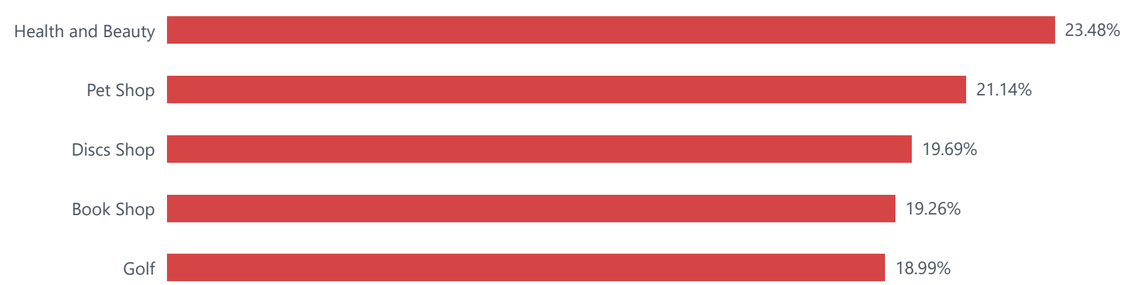
Sales by Category



Orders by Shipping Mode and Delivery Status



Where Profit Leaks ?



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SUPPLY CHAIN ANALYTICS

Delivery & Logistics Performance

Executive Overview | Delivery & Logistics... | Sales & Profitability | Customers & Geography | Data & Notes | Clear all slicers

FILTERS

Year: 2015 | 2016 | 2017 | 2018

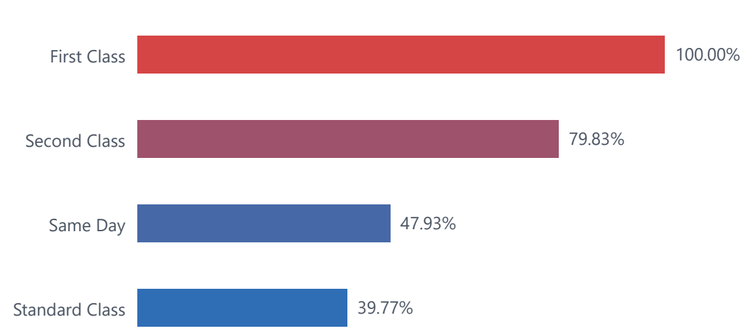
Shipping Mode: All

Customer Segment: All

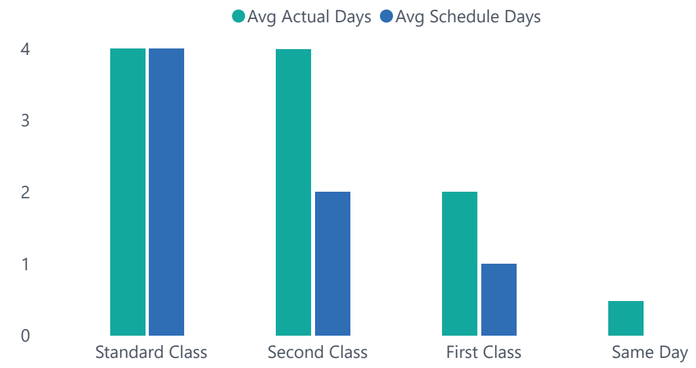
Market: All

57.29%	42.71%	3.50	2.93	0.57
Late Delivery %	On Time %	Avg Actual Days	Avg Schedule Days	Avg Delay Days

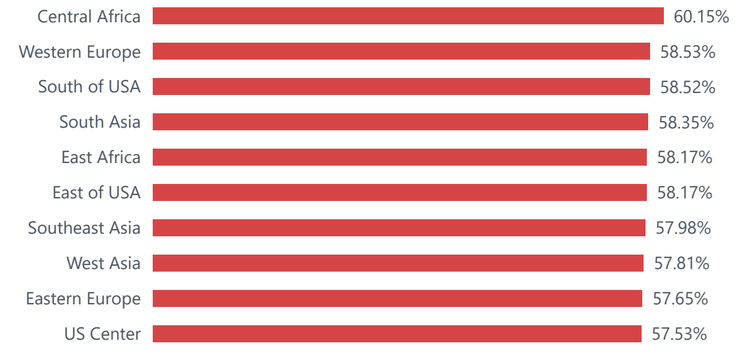
Late Delivery % by Shipping Mode



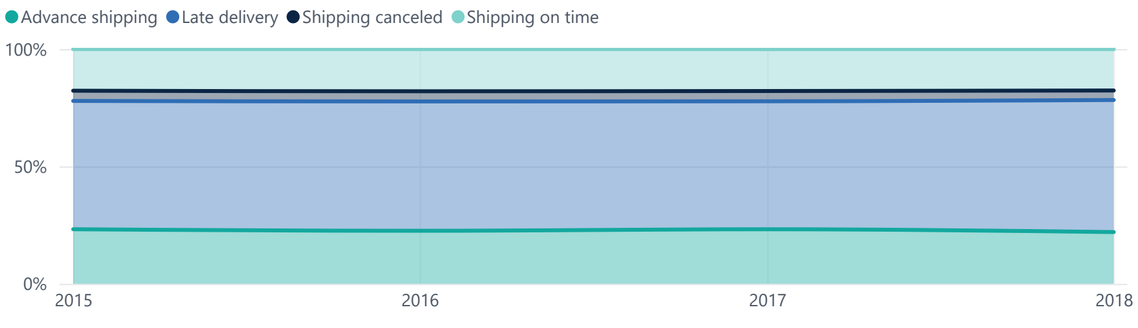
Scheduled vs Actual Days by Shipping Mode



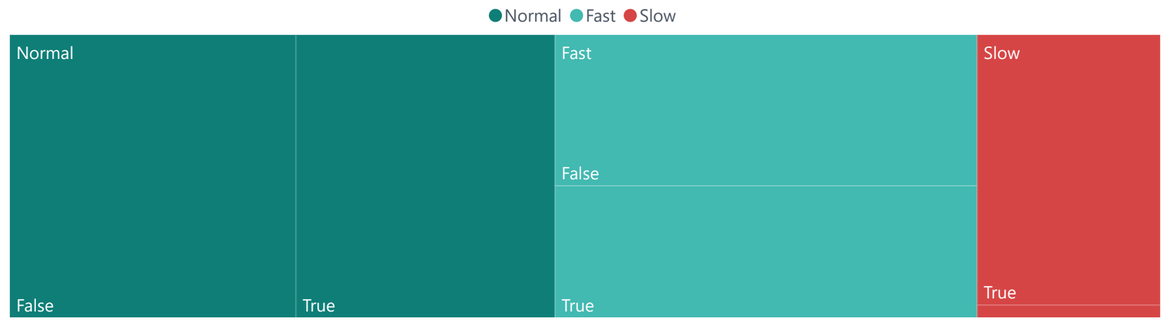
Late Delivery % by Order Region



Delivery Status trend over time



Orders by Shipping Segment and Late Delivery Risk



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SUPPLY CHAIN ANALYTICS

Sales & Profitability

FILTERS

Year: 2015 | 2016 | 2017 | 2018

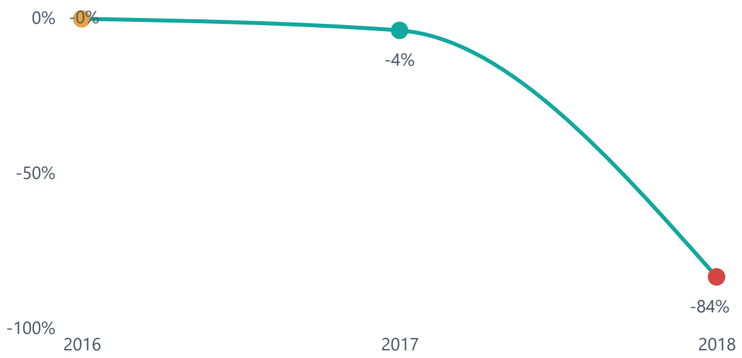
Shipping Mode: All

Customer Segment: All

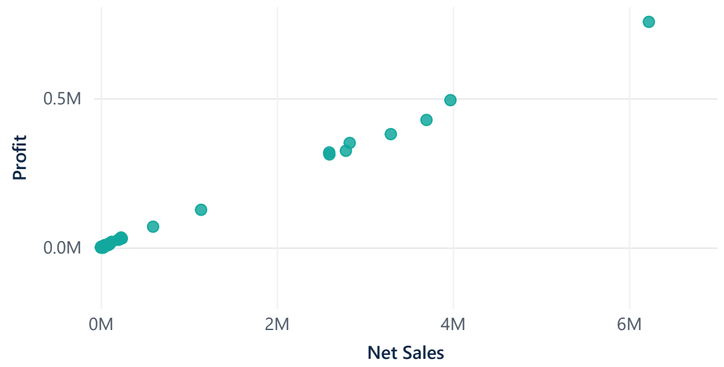
Market: All

33.05M	3.97M	12%	502.71	3.88M
Sales	Profit	Profit Margin %	Avg Order Value	Loss Amount

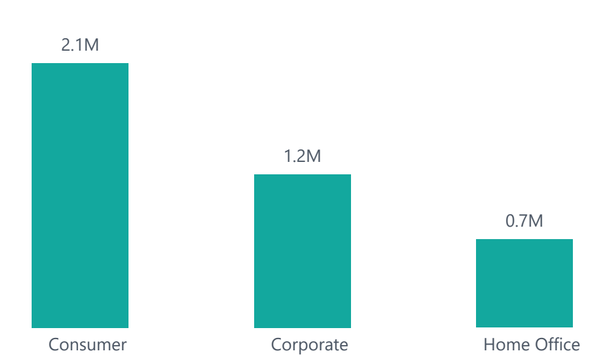
YoY Sales %



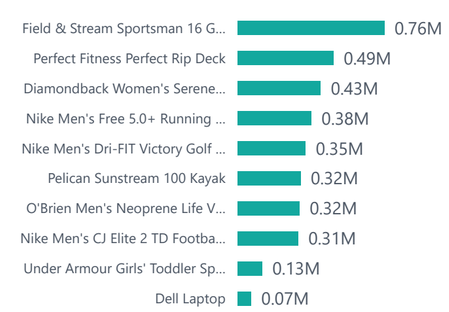
Sales vs Profit by Product



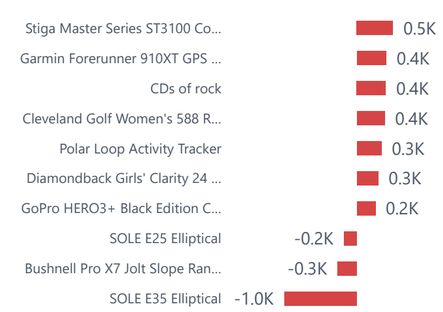
Profit by Customer Segment



Top 10 Product by Profit



Bottom 10 Product by Profit



Product Table

Product Name	Sales	Profit	Profit Margin %	Loss Making Items %
adidas Brazuca 2014 Official Match Ball	9,386.55	1,642.73	18%	15.38%
adidas Kids' F5 Messi FG Soccer Cleat	24,626.51	2,906.33	12%	16.03%
adidas Men's F10 Messi TRX FG Soccer Cleat	50,354.21	8,213.07	16%	17.70%
adidas Men's Germany Black Crest Away Tee	19,279.92	2,833.72	15%	15.92%
adidas Youth Germany Black/Red Away Match Soc	61,037.90	8,129.40	13%	22.61%
Adult dog supplies	37,318.30	3,589.26	10%	21.14%
Baby sweater	10,957.40	1,525.03	14%	16.43%
Bag Boy Beverage Holder	19,009.97	3,173.78	17%	14.34%

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SUPPLY CHAIN ANALYTICS

Customers & Geography

FILTERS

Year: 2015 | 2016 | 2017 | 2018

Shipping Mode: All

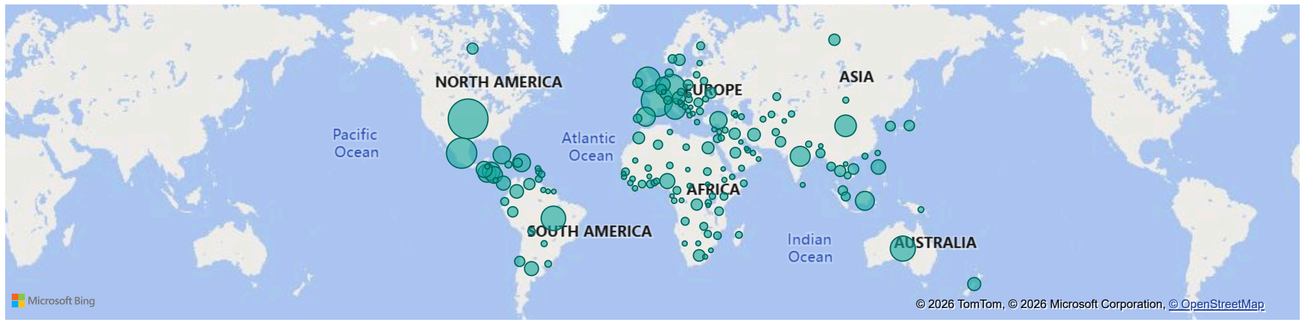
Customer Segment: All

Market: All

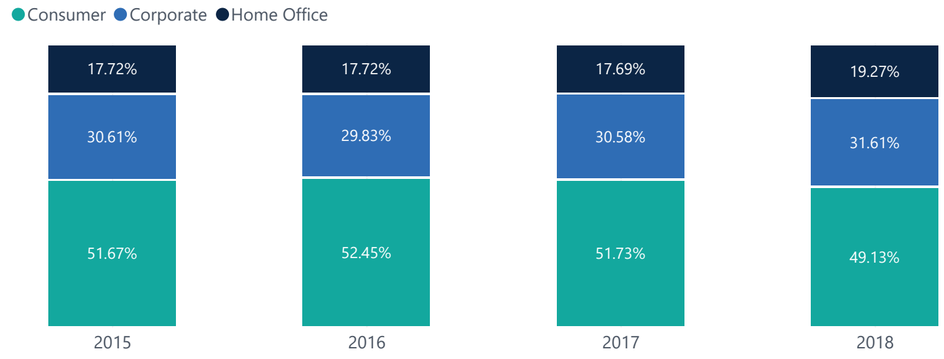
21K	3.18	1.60K	Europe
Customers	Avg Order Per Customer	Avg Sales Per Customer	Top Market

Sales by Order Country

⚠ **This visual type is being retired soon.** Upgrade now to avoid errors. Upgrade map



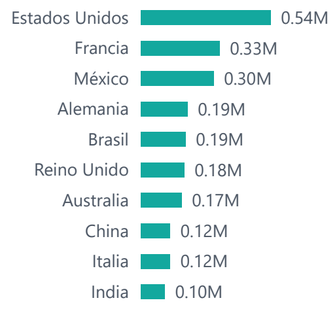
Sales by Year and Customer Segment



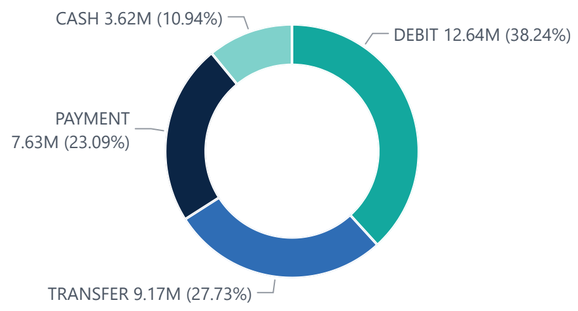
Top 10 Countries by Sales



Top 10 Countries by Profit



Sales by Transaction Type



Detail Table

Order Region	Total Orders	Sales	Late Delivery %
Canada	309	168,257.42	51.60%
Caribbean	2806	1,481,668.69	55.53%
Central Africa	556	292,912.56	60.15%
Central America	9396	5,093,849.72	57.11%
Central Asia	184	97,953.69	56.46%
East Africa	613	338,054.31	58.17%
East of USA	2323	1,231,955.13	58.17%
Eastern Asia	3318	1,334,313.41	56.72%

DATA SOURCE & PREPARATION :

- **Source**: DataCo Smart Supply Chain public dataset (DataCoSupplyChainDataset.csv), downloaded from the web link : https://www.kaggle.com/datasets/saicharankomati/dataco-supply-chain-dataset .
- **Scope**: 180,519 order lines, 65,752 orders and 20,652 customers, from January 2015 to January 2018.
- **Python (pandas)**: checked for duplicates (none) and missing values. Removed 9 redundant or mostly empty columns and converted the order and shipping dates to date type.
- **Segments created:** Profit Segment (Profit, Loss, Break Even). Shipping Segment (Fast: 2 days or less, Normal: 3–5 days, Slow: over 5 days, based on actual days). Sales Segment (four quartiles).
- **SQL Server**: Answered business questions on sales, profit, delivery performance, regions, products and customers.
- **Power BI:** Data model with a Date table and DAX measures.

Key definitions :

- **Gross Sales:** $36.78M (before discounts).
- **Net Sales:** $33.05M (Gross Sales minus $3.73M discounts).
- **Profit:** $3.97M, the sum of Order Profit Per Order.
- **Profit Margin:** Profit ÷ Net Sales, which is 12.0%.
- **Late Delivery %:** share of order lines flagged Late_delivery_risk = 1.
- **Avg Delay Days:** actual shipping days minus scheduled days.

ASSUMPTIONS & LIMITATIONS :

- **Public dataset :** The results demonstrate the analysis, not the performance of a real company.
- **2018 contains only January :** Don't compare it with full years.
- **Sales fall in November–December 2017 while orders rise :** The data doesn't explain this, so treat that period with caution.
- **Metrics are at order-line level :** One line is one product on an order. The late rate is also 54.8% when counted by whole order.
- **Cancelled and suspected-fraud orders are included in totals :** 2,855 orders, $1.41M Net Sales (4.3%).
- **Small samples by country.** 52 of 164 countries have fewer than 50 order lines, so read country-level rates with care.
- **Association, not cause.** The data has no carrier, warehouse or weather information, so it cannot explain *why* deliveries are late.
- **Gross vs Net.** The SQL scripts sum Gross Sales, and the dashboard uses Net Sales. The totals differ by the discount

INSIGHTS & RECOMMENDATIONS :

Key insights :

1. **54.8%** of order lines arrive late. Only 40.8% arrive early or on time, and 4.3% are cancelled.
2. **Faster shipping modes are the most late.** First Class is late **95.3%** of the time (promised 1 day, takes 2.0). Second Class is late 76.6% (promised 2 days, takes 4.0). Standard Class is late 38.1%.
3. **Late rates are similar in every region (about 49–58%).** That points to a process problem, not a local one.
4. **18.7%** of order lines lose money, a total of **$3.88M**. Fan Shop, Apparel and Golf account for about 80% of those losses.
5. **The business is concentrated.** Fan Shop is 46.5% of Net Sales, Consumer customers are 51.9%, and Europe plus LATAM are 57.5%.

Recommendations :

1. **Fix First and Second Class.** Renegotiate carrier terms or reset the promised days. If both matched Standard Class's late rate, late lines would fall by about **30%** (from 54.8% to about 38.5%). This is an estimate from historical rates.
2. **Review the loss-making products** in Fan Shop, Apparel and Golf, looking at price, cost and returns. Discounting alone doesn't explain the losses: the average discount rate is 10.2% on loss-making lines and 10.2% on profitable ones.
3. **Investigate the late-2017 sales drop** and the 4.3% of sales in cancelled or fraud orders.